Practice Questions

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Score: 54% Passing Score: 80%

Question 1: ✕ Incorrect

Match the network access protection (NAP) component on the left with its description on the right.

Generates a statement of health (SoH) that reports the client configuration for health requirements.

~~Enforcement server (ES)~~ NAP client

Runs the System Health Validator (SHV) program.

~~NAP client~~ NAP server

Is clients' connection point to the network.

~~NAP server~~ Enforcement server (ES)

Contain resources accessible to non-compliant computers on the limited-access network.

✓ Remediation server

EXPLANATION

NAP uses the following components:

- The NAP client generates a statement of health (SoH) that reports the client configuration for health requirements.
- A NAP server runs the System Health Validator (SHV).
- The enforcement server (ES) is clients' connection point to the network.
- The remediation server contain resources accessible to non-compliant computers on the limited-access network.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 2: ✕ Incorrect

Your company is a small start-up that has leased office space in a building shared by other businesses. All businesses share a common network infrastructure. A single switch connects all devices in the building to the router that provides internet access.

You would like to make sure that your computers are isolated from computers used by other companies. Which feature should you request to have implemented?

- VPN
- Port security
- ~~Spanning tree~~
- VLAN

EXPLANATION

Define virtual LANs (VLANs) on the switch. A port on the switch is associated with a VLAN. Only devices connected to ports that are members of the same VLAN can communicate with each other. Routers are used to allow communication between VLANs if necessary.

Use virtual private network (VPN) to securely connect two hosts through an unsecured network (such as the internet). VPN tunneling protocols protect data as it travels through the unsecured network. Spanning tree is a switch feature that allows redundant paths between switches. Port security is a method of requiring authentication before a network connection is allowed.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 3: ✔ Correct

A network switch is configured to perform the following validation checks on its ports:

- All ARP requests and responses are intercepted.
- Each intercepted request is verified to ensure that it has a valid IP-to-MAC address binding.
- If the packet has a valid binding, the switch forwards the packet to the appropriate destination.
- If the packet has an invalid binding, the switch drops the ARP packet.

Which security feature was enabled on the switch to accomplish this task?

- ➡ Dynamic ARP Inspection
- IGMP snooping
- DHCP snooping
- Port security

EXPLANATION

Dynamic ARP Inspection (DAI) is designed to prevent man-in-the middle attacks by validating ARP packets on the network. When DAI is enabled, the switch performs several validation checks on an untrusted port:

- All ARP requests and responses are intercepted.
- Each intercepted request is verified to ensure that it has a valid IP-to-MAC address binding.
- If the packet has a valid binding, the switch forwards the packet to the appropriate destination.
- If the packet has an invalid binding, the switch drops the ARP packet.

DHCP snooping filters out untrusted DHCP messages. IGMP snooping allows a switch to control which ports get IGMP traffic for a specific group. Port security restricts which hosts can connect to a switch port by MAC address.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 4: ✔ Correct

Members of the sales team use laptops to connect to the company network. While traveling, they connect their laptops to the internet through airport and hotel networks.

You are concerned that these computers will pick up viruses that could spread to your private network. You would like to implement a solution that prevents the laptops from connecting to your network unless anti-virus software and the latest operating system patches have been installed.

Which solution should you use?

- NAT
- NIDS
- ➡ NAC
- DMZ
- VLAN

EXPLANATION

Network Access Control (NAC) controls access to the network by not allowing computers to access network resources unless they meet certain predefined security requirements. Conditions that can be part of the connection requirements include requiring that computers have:

- Anti-virus software with up-to-date definition files
- An active personal firewall
- Specific operating system critical updates and patches

A client that is determined by the NAC agent to be healthy is given access to the network. An unhealthy client who has not met all the checklist requirements is either denied access or can be given restricted access to a remediation network, where remediation servers can be contacted to help the client to become compliant.

A demilitarized zone (DMZ) is a buffer network (or subnet) that sits between the private network and an untrusted network (such as the internet). A virtual LAN (VLAN) is a logical grouping of computers based on switch port. VLAN membership is configured by assigning a switch port to a VLAN. An intrusion detection system (IDS) is a special network device that can detect attacks and suspicious activity. A network-based IDS (NIDS) scans network traffic to look for intrusion attempts.

Network address translation (NAT) modifies the IP addresses in packets as they travel from one network (such as a private network) to another (such as the internet). NAT allows you to connect a private network to the internet without obtaining registered addresses for every host. Hosts on the private network share the registered IP addresses.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 5: ✕ Incorrect

A network utilizes a network access control (NAC) solution to protect against malware.

When a wired or wireless host tries to connect to the network, a NAC agent on the host checks it to make sure it has all of the latest operating system updates installed and that the latest antivirus definitions have been applied.

What is this process called?

- Quarantine
- Port security
- Posture assessment
- ~~Remediation~~

EXPLANATION

When a wired or wireless host tries to connect to the network, a NAC agent on the host checks it to make sure it has all of the latest operating system updates installed and that the latest antivirus definitions have been applied. This is called a posture assessment. The agent then submits the results of the assessment as a statement of health (SoH) to the system health validator (SHV).

If the host does not meet the client health requirements configured in the NAC system, then it is placed on a quarantine network, where it is remediated. Port security is configured on a switch to restrict connections to hosts with specific MAC addresses.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 6: ✔ Correct

Which of the following actions should you take to reduce the attack surface of a server?

- Install the latest patches and hotfixes.
- Install a host-based IDS.
- ➡ Disable unused services.
- Install anti-malware software.

EXPLANATION

Attack surface reduction (ASR) cuts down on the software or services running on a system. By removing unnecessary software, features, or services, you eliminate possible attacks directed at those components. When removing unnecessary components:

- Use role separation by installing services on separate physical systems. If a single system is compromised, only the few services on that system will be affected.
- For many new systems, unnecessary services are often installed by default. Following installation, you should remove unneeded services, protocols, and applications.
- When removing existing services, determine the unneeded services and their dependencies before altering the system.

Adding anti-malware or a host-based IDS adds a level of protection (defense in depth), but does not reduce the number of components running on the system. Applying patches is necessary to fix security problems with software or the operating system, but if the system is not running a specific piece of software, the patches that apply to that software are irrelevant and will not need to be applied.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 7: ✔ Correct

Which type of security uses MAC addresses to identity devices that are allowed or denied a connection to a switch?

- ➡ Port security
- Secure Sockets Layer
- MAC spoofing
- Traffic shaping

EXPLANATION

Port security uses the MAC address to identify allowed and denied devices.

Traffic shaping is the practice of modifying the flow of data through a network to assure a certain level of performance. Secure Sockets Layer (SSL) is a protocol used to secure IP protocol traffic. MAC spoofing is a technique attackers use to gain access to a network that uses MAC filtering.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 8: ✔ Correct

Match the port security MAC address type on the left with its description on the right.

A MAC address manually identified as an allowed address.

✔ SecureConfigured

A MAC address that has been learned and allowed by the switch.

✔ SecureDynamic

A MAC address that is manually configured or dynamically learned that is saved in the config file.

✔ SecureSticky

EXPLANATION

MAC addresses are stored in RAM in the CAM table and are identified with the port and by a MAC address type. Port security uses the following three MAC address types:

Type	Description
SecureConfigured	A SecureConfigured address is a MAC address that has been manually identified as an allowed address.
SecureDynamic	A SecureDynamic address is a MAC address that has been dynamically learned and allowed by the switch.
SecureSticky	A SecureSticky address is a MAC address that is manually configured or dynamically learned and saved.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 9: ✕ Incorrect

You manage a network that uses switches. In the lobby of your building are three RJ45 ports connected to a switch.

You want to make sure that visitors cannot plug in their computers into the free network jacks and connect to the network, but you want employees who plug into those same jacks should be able to connect to the network.

What feature should you configure?

- VLANs
- ➡ Port authentication
- Mirroring
- ~~Spanning tree~~
- Bonding

EXPLANATION

Use port authentication to prevent unauthorized access through switch ports. Port authentication is provided by the 802.1x protocol and allows only authenticated devices to connect to the LAN through the switch. Authentication uses usernames and passwords, smart cards, or other authentication methods.

- When a device first connects, the port is set to an unauthorized state. Ports in unauthorized states can only be used for 802.1x authentication traffic.
- After the server authenticates the device or the user, the switch port is placed in an authorized state, and access to other LAN devices is allowed.

With a VLAN, you assign each port to a VLAN. If the ports in the lobby were assigned to one VLAN, you could control the type of access through the switch for those ports, but could not modify the access based on user. If you use a VLAN, both visitors and employees would have the same access through those ports.

Spanning tree is a protocol on a switch that allows the switch to maintain multiple paths between switches within a subnet. The spanning tree protocol runs on each switch and is used to select a single path between any two switches. Mirroring sends traffic from all switch ports to a switch port you designate as the mirrored port. Bonding allows multiple switch ports to be used at the same time to reach a specific destination.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 10: ✕ Incorrect

In which of the following situations would you use port security?

- ~~You want to prevent MAC address spoofing.~~
- You want to control the packets sent and received by a router.
- ➡ You want to restrict the devices that could connect through a switch port.
- You want to prevent sniffing attacks on the network.

EXPLANATION

Use port security on a switch to restrict the devices that can connect to a switch. Port security uses the MAC address to identify allowed and denied devices. When an incoming frame is received, the switch examines the source MAC address to decide whether to forward or drop the frame.

Port security cannot prevent sniffing or MAC address spoofing attacks. Use an access list on a router to control sent and received packets.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 11: ✕ Incorrect

You are the network administrator for a city library. Throughout the library, there are several groups of computers that provide public access to the internet. Supervision of these computers has been difficult. You've had problems with patrons bringing personal laptops into the library and disconnecting the network cables from the library computers to connect their laptops to the internet.

The library computers are in groups of four. Each group of four computers is connected to a hub that is connected to the library network through an access port on a switch. You want to restrict access to the network so only the library computers are permitted connectivity to the internet.

What can you do to fix this problem?

- Configure port security on the switch.
- Create static MAC addresses for each computer and associate them with a VLAN.
- Create a VLAN for each group of four computers.
- ~~Remove the hub and place each library computer on its own access port.~~

EXPLANATION

Configuring port security on the switch can restrict access so that only specific MAC addresses can connect to the configured switch port. This would prevent the laptop computers from being permitted connectivity.

Placing each library computer on its own access port would have no affect.

VLANs are used to group broadcast traffic and do not restrict device connectivity as this scenario requires.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 12: ✔ Correct

You have a company network with a single switch. All devices connect to the network through the switch.

You want to control which devices will be able to connect to your network. For devices that do not have the latest operating system patches, you want to prevent access to all network devices except for a special server that holds the patches that the computers need to download.

Which of the following components will be part of your solution? (Select two.)

- [ ] DMZ
- [x] 802.1x authentication
- [ ] Honeypot
- [ ] Extranet
- [x] Remediation servers

EXPLANATION

Network access control (NAC) controls access to the network by not allowing computers to access network resources unless they meet certain predefined security requirements. NAC can be used with 802.1x port authentication on a switch to allow or deny access to the network through the switch port. A client that is determined by the NAC agent to be healthy is given access to the network. An unhealthy client who has not met all the checklist requirements is either denied access or given restricted access to a remediation network, where remediation servers can help the client to become compliant. For example, remediation servers might include anti-virus software and definition files that can be installed. If and when the unhealthy client's status changes to healthy, the client is given access to the network.

A demilitarized zone (DMZ) is a buffer network (or subnet) that sits between the private network and an untrusted network (such as the internet). DMZs are created with routers and firewall rules to allow or block traffic. DMZs use information in the packet to allow or deny packets.

An extranet is a privately-controlled network that is distinct from but located between the internet and a private LAN. An extranet is often used to grant resource access to business partners, suppliers, and even customers outside of the organization. A honeypot is a device or virtual machine that entices intruders by displaying a vulnerable trait or flaw or by appearing to contain valuable data.

REFERENCES

LabSim for Network Pro, Section 14.3.

Question 13: ✔ Correct

A network switch detects a DHCP frame on the LAN that appears to have come from a DHCP server that is not located on the local network. In fact, it appears to have originated from outside the organization's firewall.

As a result, the switch drops the DHCP message from that server.

Which security feature was enabled on the switch to accomplish this?

- Dynamic ARP inspection
- ➡ DHCP snooping
- Port security
- IGMP snooping

EXPLANATION

Some switches provide DHCP snooping as a security feature. DHCP snooping filters untrusted DHCP messages. An untrusted DHCP message is received from outside the network or firewall. DHCP snooping acts like a firewall between DHCP clients and your DHCP servers. The switch maintains a DHCP snooping binding table that matches MAC addresses with DHCP messages. When DHCP snooping is enabled, the switch drops DHCP messages if the frame from the DHCP server is received from outside the network or firewall. It also drops DHCP messages if the source MAC address and the DHCP client MAC address do not match in the DHCP snooping binding table.

IGMP snooping allows a switch to control which ports get IGMP traffic for a specific group. Port security restricts which hosts can connect to a switch port based on MAC addresses. Dynamic ARP Inspection is designed to prevent man-in-the middle attacks by validating ARP packets on the network.

REFERENCES

LabSim for Network Pro, Section 14.3.

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